Hari krishna V
251901050

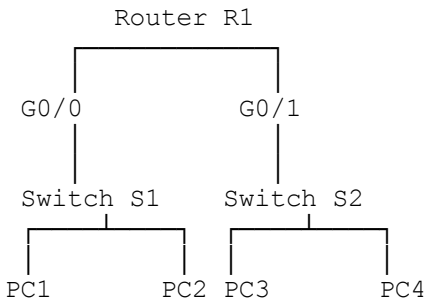
COMPUTER NETWORKS

Ex.No.10	**Design a simple topology and configure with one router, two switches and PCs using Cisco Packet Tracer**
DATE:	

AIM: To design and configure a simple network topology with one router, two switches, and four PCs in Cisco Packet Tracer for successful inter-network communication.

PROCEDURE:

1. Topology



2. Devices to place

In **Cisco Packet Tracer**, place:

- **1 Router** – Cisco 1941
- **2 Switches** – Cisco 2960
- **4 PCs**

Use **Copper Straight-Through** cables.

3. IP Addressing

Device	Interface	IP Address	Subnet Mask	Gateway
R1	G0/0	192.168.1.1	255.255.255.0	—
R1	G0/1	192.168.2.1	255.255.255.0	—
PC1	NIC	192.168.1.10	255.255.255.0	192.168.1.1
PC2	NIC	192.168.1.11	255.255.255.0	192.168.1.1
PC3	NIC	192.168.2.10	255.255.255.0	192.168.2.1
PC4	NIC	192.168.2.11	255.255.255.0	192.168.2.1

So we have **two different networks**:

```
Network 1: 192.168.1.0/24
       PC1 ─┐
       PC2 ─┤
            S1 ── R1 G0/0 = 192.168.1.1

Network 2: 192.168.2.0/24
       PC3 ─┐
       PC4 ─┤
            S2 ── R1 G0/1 = 192.168.2.1
```

4. Configure Router R1

Click **R1 → CLI** and enter:

```
enable
configure terminal

interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

interface gigabitEthernet 0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
```

```
exit

end
write memory
```

The important command is:

```
no shutdown
```

It activates the router interface.

Check the interfaces using:

```
show ip interface brief
```

You should see the interfaces as **up/up**.

5. Configure PC1

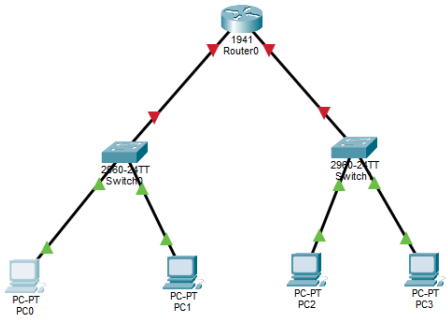
Go to:

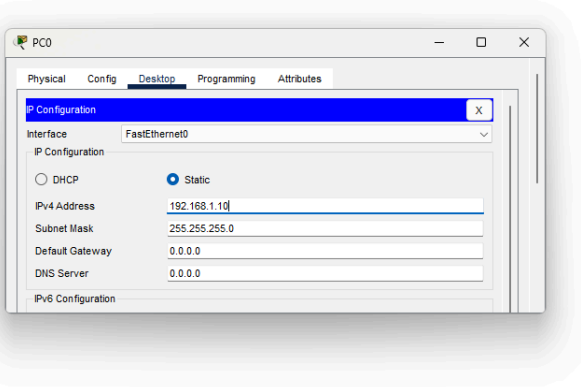
PC1 → Desktop → IP Configuration

Enter:

```
IP Address:       192.168.1.10
Subnet Mask:      255.255.255.0
Default Gateway:  192.168.1.1
```

SCREENSHOTS:(to be pasted)

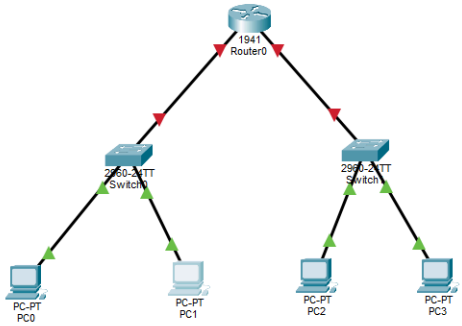


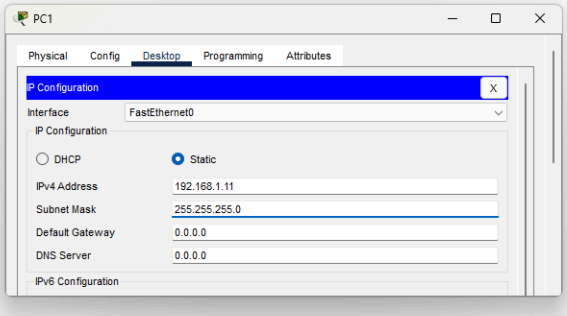


6. Configure PC2

```
IP Address:        192.168.1.11
Subnet Mask:       255.255.255.0
Default Gateway:   192.168.1.1
```

SCREENSHOTS:(to be pasted)

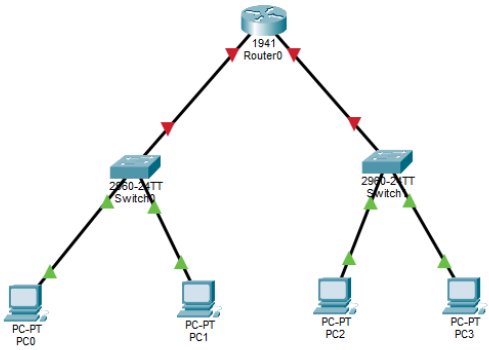


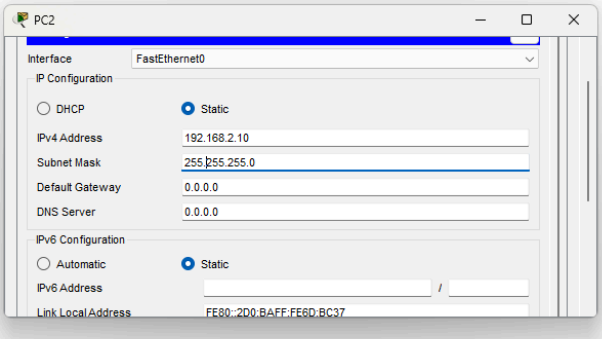


7. Configure PC3

```
IP Address:        192.168.2.10
Subnet Mask:       255.255.255.0
Default Gateway:   192.168.2.1
```

SCREENSHOTS:(to be pasted)

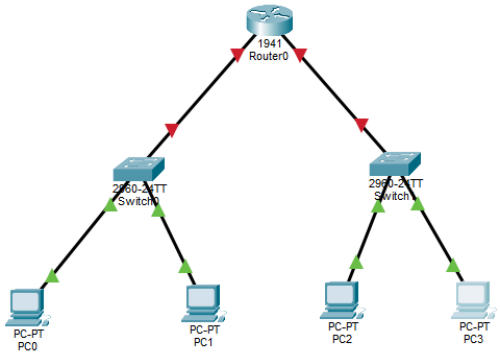


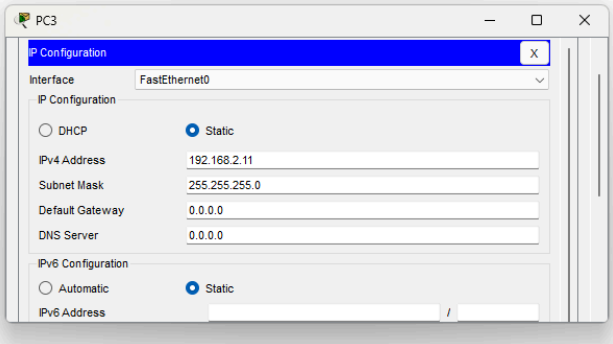


8. Configure PC4

```
IP Address:       192.168.2.11
Subnet Mask:      255.255.255.0
Default Gateway:  192.168.2.1
```

SCREENSHOTS:(to be pasted)





9. Test the network

From **PC1 → Desktop → Command Prompt**, type:

```
ping 192.168.1.11
```

This tests PC1 → PC2.

Then:

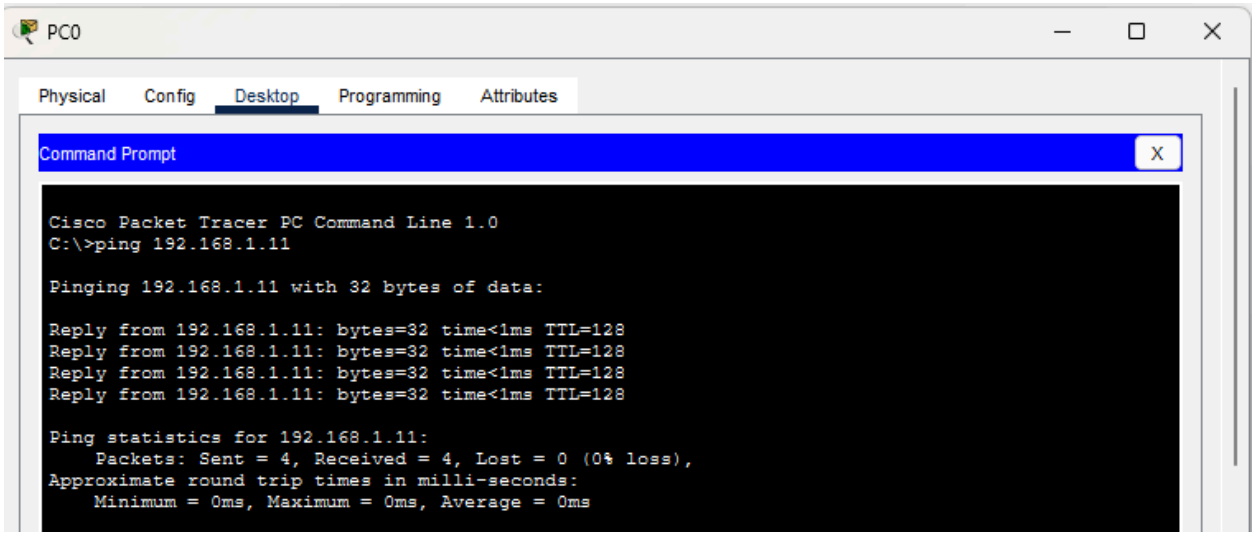
```
ping 192.168.2.10
```

This tests **PC1 → Router → Switch 2 → PC3**.

You should receive:

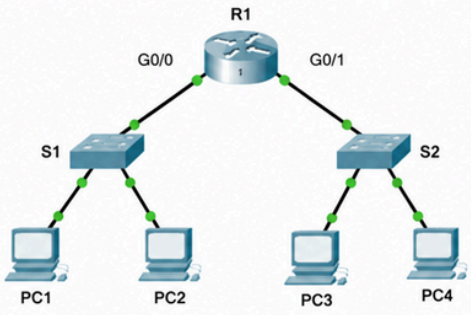
```
Reply from 192.168.2.10
Reply from 192.168.2.10
Reply from 192.168.2.10
Reply from 192.168.2.10
```

SCREENSHOTS:(to be pasted)



SAMPLE CONFIGURATION:

TOPOLOGY: One Router, Two Switches and PCs



IP ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	192.168.1.1	255.255.255.0	-
R1	G0/1	192.168.2.1	255.255.255.0	-
PC1	NIC	192.168.1.10	255.255.255.0	192.168.1.1
PC2	NIC	192.168.1.11	255.255.255.0	192.168.1.1
PC3	NIC	192.168.2.10	255.255.255.0	192.168.2.1
PC4	NIC	192.168.2.11	255.168.255.0	192.168.2.1

CONFIGURATION STEPS

1. Place Devices

- 1 Router (e.g., 1941)
- 2 Switches (e.g., 2960)
- 4 PCs
- Connections: Copper Straight-Through

2. Configure Router (R1)

```
R1> enable
R1# configure terminal
R1(config)# interface g0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface g0/1
R1(config-if)# ip address 192.168.2.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# end
R1# write memory
```

3. Configure PCs

For PC1

- IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

For PC2

- IP Address: 192.168.1.11
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

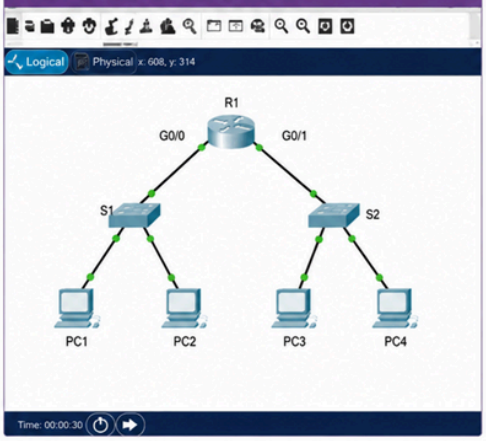
For PC3

- IP Address: 192.168.2.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.1

For PC4

- IP Address: 192.168.2.11
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.1

PACKET TRACER TOPOLOGY VIEW



VERIFY CONNECTIVITY

From any PC, open Command Prompt and ping other PCs in the same network and across networks.
Example: From PC1 ping PC3 → `ping 192.168.2.10`
All pings should be successful.

RESULT: Therefore the process to design and configure a simple network topology with one router, two switches, and four PCs in Cisco Packet Tracer for successful inter-network communication has been simulated, studied and verified.